

CYLINDER DISTRIBUTION PLATFORM.

Production-Grade LPG Delivery Backend

TEAM CYLDIST

Department of Computer Engineering

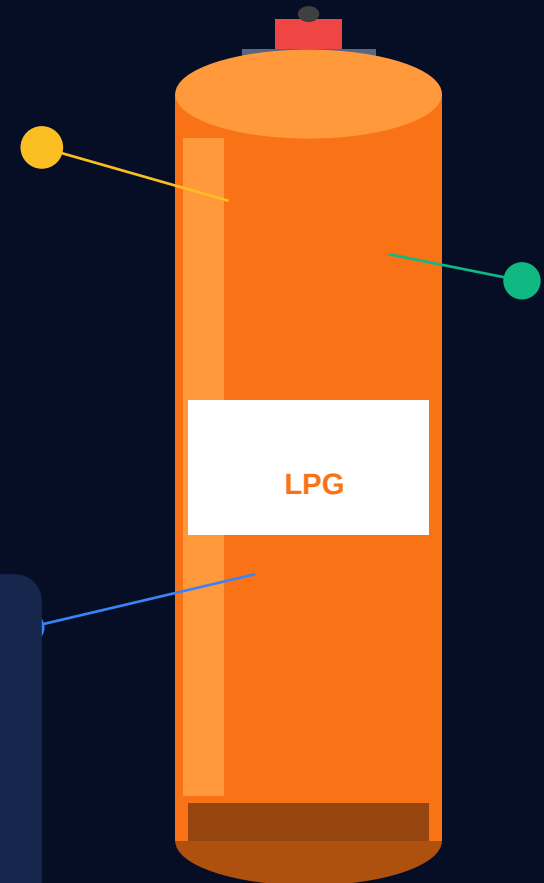
Member 1 - Lead Developer

Member 2 - Backend Engineer

Member 3 - DevOps & DB

Member 4 - Frontend & QA

[Your College Name] | Academic Year 2025-26



02

INTRODUCTION

Digital LPG Distribution Reimagined.

A production-grade backend that transforms how LPG cylinders are booked, tracked, and delivered.

PROJECT IMPACT

3+

User Roles

10+

Core Features

30+

API Endpoints

2

Databases



KEY OBJECTIVES

Goals & Mission

1

DIGITIZE

Replace manual booking
with online + WhatsApp

2

TRACK

Real-time GPS tracking
& live customer chat

3

SECURE

JWT + RBAC + bcrypt
production security

4

SCALE

Docker + Kubernetes
ready deployment

5

CRISIS ENGINE

Emergency-aware allocation with Priority Score Formula

HOSPITAL +150

DOMESTIC +75

HOTEL 70% cap

HOARDING -200

Problem vs Solution



TRADITIONAL SYSTEM

Outdated - Manual - Inefficient

- ✗ Manual booking via phone calls
- ✗ No real-time delivery tracking
- ✗ Opaque inventory management
- ✗ Zero priority for emergencies
- ✗ Paper-based record keeping
- ✗ No customer communication
- ✗ Inefficient route planning



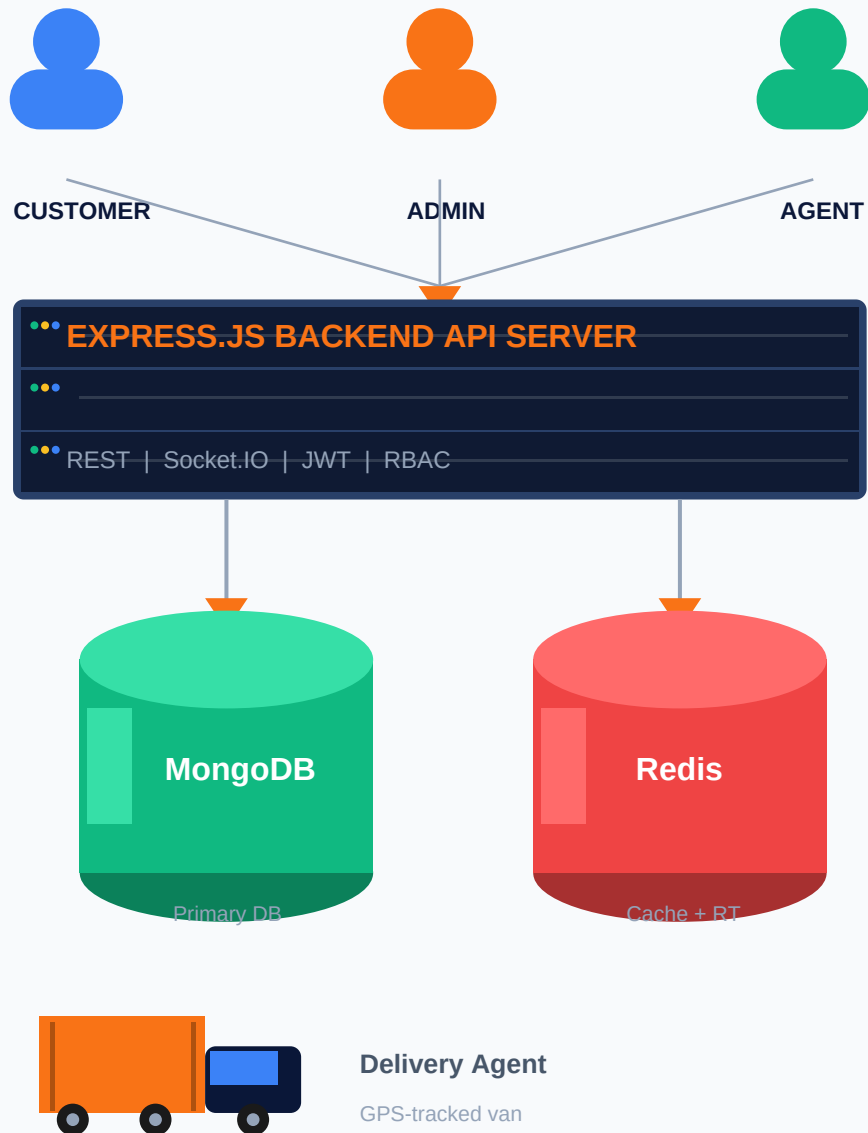
OUR DIGITAL SOLUTION

Modern - Automated - Scalable

- ✓ Online + WhatsApp booking
- ✓ Live GPS delivery tracking
- ✓ Automated inventory engine
- ✓ Crisis prioritization system
- ✓ Digital order management
- ✓ Real-time agent-customer chat
- ✓ Smart route optimization

Architecture & Workflow

ARCHITECTURE FLOW



ORDER LIFECYCLE FLOW

- 01 Customer places order
- 02 Admin assigns agent
- 03 Agent picks cylinder
- 04 GPS tracking begins
- 05 Live chat enabled
- 06 Delivery + OTP verify
- 07 Order completed!

How Emergency Allocation Works

During severe stock crises, FCFS is suspended. Orders enter a holding pool & are allocated using a Heuristic Priority Score (P).

PRIORITY SCORE FORMULA

$$P = (1.5 \times S_sector) + (1.0 \times S_urgency) - (1.0 \times S_hoarding)$$

Higher P = higher priority. Orders sorted in a Max-Heap and allocated in descending P order.

S_sector

Sector Base Score

Hospitals/Emergency: 100
Domestic Households: 50
Hostels/Institutional: 30
Commercial (Hotels): 10

Multiplier: $W_sector = 1.5$
Hospital order = 150 in P

S_urgency

Necessity Index

$(DaysSinceRefill / AvgCycle)$
x 100

Capped at 200 max
(prevents outlier anomalies)

Multiplier: $W_urgency = 1.0$

S_hoarding

Hoarding Penalty Shield

-200 if user refilled within
21 days, OR ordered more
than 2 cylinders in 30 days

Hospitals: fully exempt

Multiplier: $W_hoarding = 1.0$

HOSPITAL

Exempt from cooldowns

15% emergency reserve stock

DOMESTIC HOUSEHOLD

30-day lock period

Hard reject within 30 days

HOTEL / COMMERCIAL

7-day lock period

70% quantity cap applied

Order Created



Score Calculated



Max-Heap Sort



Batch Allocate



Notify Customer

06

TECH STACK

Modern Technologies

3D-illustrated stack for production scalability



SUPPORTING LIBRARIES & TOOLS

Security: bcrypt | Helmet | CORS | Zod

Integration: Twilio | SendGrid | Razorpay | S3

Logging: Winston | Morgan | rotate-file

Testing: Jest | Supertest | mongo-mem-server

Unique Features

01

WhatsApp Booking

API-based booking via WhatsApp

02

Product Store

Buy accessories with cylinders

03

Live GPS Tracking

5-second GPS update intervals

04

Real-time Chat

Customer-Agent messaging

05

Crisis Engine

Emergency-aware allocation

06

Inventory Manager

Warehouse-level cylinder tracking

07

JWT + RBAC

Role-based access control

08

Multi-channel Notify

SMS / Email / Push alerts

09

Swagger API Docs

Interactive API testing UI

10

Docker + K8s

Production-grade deployment

11

DELIVERY MANAGEMENT

End-to-end delivery lifecycle: assignment, tracking, verification & proof of delivery

Real-time GPS via Socket.IO | `/delivery/:orderId/route` API | OTP verify @ delivery

Agent Assign

Live GPS

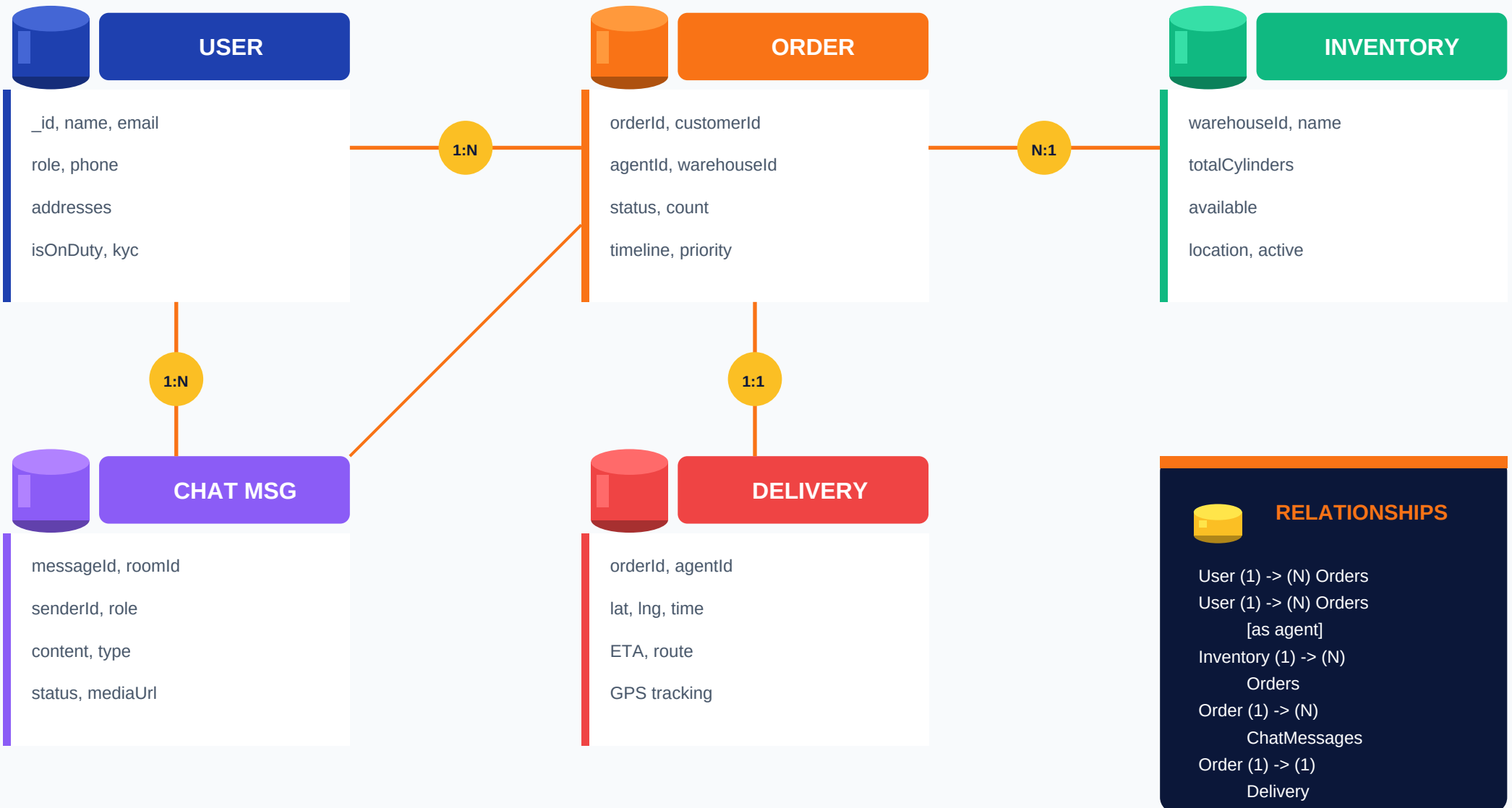
OTP Verify

Photo Proof

5-second GPS interval | Socket.IO rooms per order | Order timeline tracking

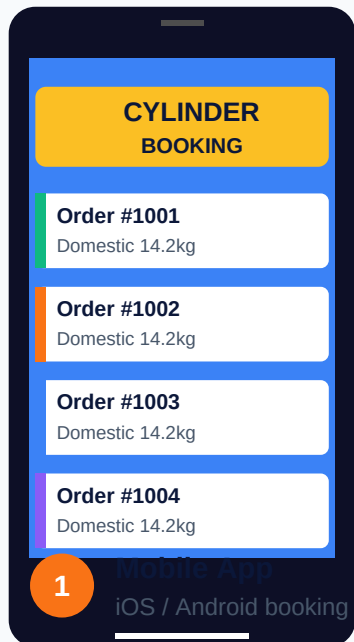
Delivery API endpoints + admin/agent routing dashboard

Database Design & ER Diagram



Demo & Security

EXECUTION DEMO



1

Mobile App
iOS / Android booking

2

Admin Panel

Order management UI

3

Live GPS

Real-time tracking

4

API Docs

Swagger interactive UI



SECURITY ARCHITECTURE

**MULTI-LAYER PROTECTION**

Production-grade

JWT AUTHAccess (15min) + Refresh (7d)
Token rotation + reuse detect**RATE LIMITING**100 req/15min global
10 req/15min for auth**ENCRYPTION**bcrypt 12-round hashing
Helmet + CSP + HTTPS**RBAC**Admin | Customer | Agent
Role-scoped routes

Helmet

CORS

HPP

Sanitize

XSS

Winston

Conclusion & Future Scope

PROJECT ACHIEVEMENTS

- ✓ Production-grade REST API
- ✓ Real-time GPS + Live Chat
- ✓ Crisis Prioritization Engine
- ✓ Complete Order Lifecycle
- ✓ Docker & Kubernetes Deploy
- ✓ Multi-layer Security (JWT/RBAC)
- ✓ WhatsApp Integration
- ✓ Swagger API Documentation
- ✓ Inventory Management
- ✓ Real-time Notifications

FUTURE ENHANCEMENTS

- AI**
Demand Prediction
ML-based consumption forecast
- APP**
Mobile App
React Native / Flutter
- PAY**
Online Payments
Razorpay / Stripe / UPI
- MAP**
Route Optimization
Google Maps integration
- IoT**
Smart Sensors
Cylinder gas-level IoT

THANK YOU!

Questions & Discussion Welcome

Zonal Crisis Prioritization & Heuristic Allocation Engine

Objective: Allocate LPG fairly during severe shortages by replacing FCFS with a priority-based heuristic system

[1] CRISIS MODE ACTIVATION

- FCFS allocation is suspended during shortages
- Incoming orders enter a Zonal Holding Pool
- Orders periodically ranked by Priority Score (P)
- Allocation runs in batches via Max-Heap engine

[2] ALLOCATION WORKFLOW



Highest Priority Score served first - Max-Heap descending

Periodic batch allocation cycles ensure fairness across the zone

[3] PRIORITY SCORE FORMULA

$$P = (1.5 \times S_{\text{sector}}) + (1.0 \times S_{\text{urgency}}) - (1.0 \times S_{\text{hoarding}})$$

S_{sector} : Hospital 100 / Domestic 50 / Hostel 30 / Commercial 10 | S_{urgency} : (DaysSinceRefill / AvgCycle) x 100, max 200 | S_{hoarding} : -200 if refill <21d (Hospital exempt)

[4] HOSPITAL & EMERGENCY

Highest Priority

- Dedicated emergency reserve
- No cooldown restrictions
- Exempt from hoarding shield

[4] DOMESTIC HOUSEHOLD

30-day Cooldown

- Hard reject within 30 days
- Anti-hoarding lock active
- Standard priority weight

[4] COMMERCIAL / HOTELS

7-day + 70% Cap

- 7-day booking cooldown
- 70% quantity reduction cap
- Lowest priority weight

[5] KEY BENEFITS

- ✓ Prevents hoarding
- ✓ Fair distribution
- ✓ Emergency-first
- ✓ Necessity-based
- ✓ Better utilization

Outcome: Data-driven heuristic ensuring equitable & efficient LPG distribution during emergencies

System Architecture & Design

Three-tier architecture: Client Layer -> API Gateway -> Data + External Services

[TIER 1] CLIENT LAYER - React 19

CUSTOMER APP

Book - Track - Chat - Rate

ADMIN PANEL

Dispatch - Crisis - Analytics

AGENT APP

GPS - Photo Proof - Chat

HTTP REST

WebSocket (Socket.IO)

[TIER 2] API GATEWAY - Node.js + Express + Socket.IO

Auth

Orders

Dispatch

Crisis

Chat

Inventory

Delivery

Notify

Support

MIDDLEWARE CHAIN:

Helmet | CORS | Rate Limiter (100/15min) | JWT Auth (15m+7d) | RBAC | Mongo Sanitize | XSS Clean | HPP

[TIER 3] DATA & EXTERNAL SERVICES

MongoDB

Primary DB - Users, Orders, Inventory

Redis

Cache, GPS, Sessions, OTP, Rate Limits

External APIs

Ola Maps, Razorpay, AWS S3, SMTP, SMS

[*] PERFORMANCE METRICS

Auto-Dispatch (9 ord)	55-90 ms
Redis ops / sec	6,289 reads
API (cached)	< 5 ms
GPS broadcast	every 5 s
Concurrent sockets	1000+

[*] CORE ALGORITHMS

K-Means++	Order GPS clustering
Greedy Nearest	Agent-cluster pairing
Priority-Weighted	Route sequencing
Heuristic P-Score	Crisis allocation
Max-Heap Ranking	Descending P order

[*] ORDER LIFECYCLE

- 1 CREATED
- 2 ASSIGNED
- 3 OUT FOR DELIVERY
- 4 DELIVERED

OTP + Photo proof verification at delivery

Production-grade backend - Docker + K8s ready - REST + WebSocket - Multi-database (MongoDB + Redis)